

Fig. 2: Functional diagram of Toffoli gate

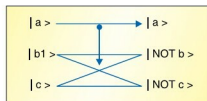


Fig. 3: Functional diagram of controlled swapped gate

TABLE V
Truth Table of CNOT Gate

Input 1	Input 2	Output 1	Output 2
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

wise, these are left as it is. That's why Fredkin gate is called a swapped gate.

Another type of a controlled swapped gate is one that has input and output relationship follows:

For inputs of a, b, c

Outputs = a, b, c when a=0

Outputs = a, c, b when a=1

The mapping of this type of controlled swapped gate is shown in Table IV. Functional diagram is shown in Fig. 3.

Controlled NOT gate. This is two-input, two-output gate with truth table as shown in Table V. The first output is the copy of the first input. The second output is NOT of the second input if the first input is 1. Otherwise,

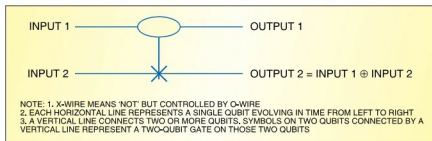


Fig. 4: CNOTIXOR gate

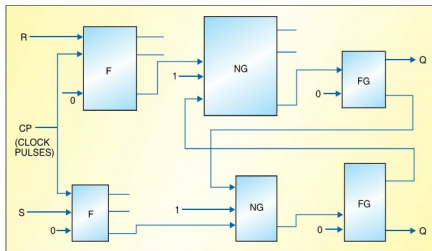


Fig. 5: Proposed reversible RS flip-flop

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The CNOT gate may be explained from different angles to make it more analogous to the classical gate. There are infinite numbers of single-qubit quantum gates, unlike only two (identity and logical NOT) in classical information. The quantum NOT gate performs $|0\rangle$ to $|1\rangle$ and vice versa analogous to classical NOT. Thus CNOT gate can be defined as:

1. The second qubit undergoes NOT if and only if the first qubit is in state $|1\rangle$

2. The effect of CNOT may be written as:

First output = first input

Second output = first input $\oplus$ second input

For this reason, CNOT gate is also called XOR gate. CNOT gate is shown in Fig. 4.

Other logical operations require additional qubits. The most popular three-qubit gate is controlled-controlled NOT gate, or CCN or C²NOT gate. This gate is also known as Toffoli gate—named after Toffoli who demonstrated that the classical version is universal for classical reversible computation.

RS flip-flop. Lots of combinational quantum gates have been made till date. Sequential gates are yet to take off in a big way. A few sequential quantum gates are theoretically reported by some researchers. A proposed quantum RS flip-flop is shown in Fig. 5. This gate is made of Fredkin gate and a new gate.

Success so far

It is not yet possible to conceive all the applications of mature quantum computers. The largest number of qubits ever contained within a quantum computing system is seven. As quantum computing research furthers rapidly with funding of many millions of dollars, it will only be a matter of time before a critical breakthrough occurs and impressive applications are invented. ●

The author is a regular contributor to EPT